Model 3 — daily triage: DESIGN SPECIFICATION (v2)

SADINhub Energy Intelligence · DIOM Facility, Riyadh · Model 3 v1 IMPLEMENTED · 14,130 scored meter-days · 175 tests

Status: design only, approved in principle 2026-09-20 with five decisions resolved and four corrections applied. No Isolation Forest is fitted, no score is produced, no threshold is chosen. Every number was measured read-only against the live database from `gold.energy_raw` + `gold.meter` alone, with no dependency on Task 0A or Task 0B.

Revision v2 applies corrections A–D of 2026-09-20 — cross-refit ranking, artifact separation from Model 1, corrected feature-range assumptions, and base rates recomputed on the real scored universe. Each is marked **[A]**, **[B]**, **[C]**, **[D]** where it lands.

---

0. What Model 3 is, and what it is not

Is	A secondary daily triage score that ranks meter-days by how unusual their raw electrical behaviour was, so limited engineering review time goes to the most unusual days first.
Is not	The primary anomaly detector. That is Model 1 — deterministic, evidence-carrying, frozen.
Is not	A ground-truth classifier. No engineering labels exist; nothing here establishes that a day contains a fault.
Is not	A consumption model. It reads no register-derived quantity and does not depend on Task 0A or 0B.

Model 3 produces a **ranking**, never a verdict.

---

1. The ten decisions

Decision 1 — Zone 13 stays in scope. 31 meters.

Model 3 uses **no register-derived feature**, so Zone 13's cumulative-kWh integrity concern does not reach any input it consumes.

	Zone 13 (2 meters)	Other 29
Median readings/day	269	287
Median **ungated** PF-Critical rate, phase 1	**0.3204**	**0.0105**
Median **load-gated** PF-Critical rate, phase 1	**0.0000**	**0.0000**
Days the 5% CB gate opens	83.4%	72.2%

The 5.6% cadence difference is far smaller than the 23.8% shift the whole facility underwent over time (§3), which rates already neutralise. The 30× PF gap **disappears entirely under load gating** — it was measuring how lightly loaded those meters run.

> Conditional on the PF feature being load-gated (Decision 2). Ungated, Zone 13's two meters would look permanently anomalous and consume review capacity every day for a non-issue.

Decision 2 — exact numerator and denominator for every rate

Denominator	Meaning	Used by
`R` = `readings_reported`	reading packets that meter produced that Saudi day	per-reading and per-phase-channel rates
`3R`	phase-channels, three per reading	rates pooled across the three phases
`G_k` = `gated_readings_phase_k`	readings where phase *k* is ≥ 5% of breaker rating **and** carries forward power	the load-gated power-factor rate

Full definitions in §4. Every **rate** is a proportion whose numerator is a subset of its denominator, so every rate is in [0, 1] and is unchanged if the meter reports twice as often. Features that are *not* rates are not so bounded — see Decision 11 **[C]**.

Decision 3 — partial and missing days

Case	Treatment
No readings at all	No row. A day with no telemetry has no electrical behaviour to rank. Absence is Model 1 T4's `data_availability` job, already detected there.
Before a meter's first reading / after its last	Excluded — outside the meter's life, not a gap.
Interior gap day	No row; counted as `missing_interior_days` in run metadata.
Partial day (few readings)	**Retained**, with `coverage_ratio` — Decision 4.

Measured: **15,759 meter-days exist** of 17,980 possible (580 × 31), so **2,221 are absent**. Largest contributors: ZN.17-PB (227 interior gap days, its 201-day outage) and ZN.9-PB (215). Fourteen calendar dates have **no meter reporting at all**.

Decision 4 — low-reading-count days are retained, with a coverage feature

1. Exclusion needs an arbitrary threshold and the distribution offers no natural cut: 0.22% of meter-days are below 10 readings, 0.47% below 50, 1.15% below 200.
2. The thinnest days are often the most interesting.
3. A rate handles cadence; what it does not convey is *confidence*, which is what coverage carries.

```
coverage_ratio = readings_reported / median(readings_reported over all meters that Saudi date)
```

Data-derived, so it absorbs the 286 → 354 cadence regime change automatically and survives any one meter being down. `low_coverage_day` (`coverage_ratio < 0.5`) is carried **for reporting only** and is never a filter.

Decision 5 — one global Isolation Forest, not one per meter

	Global	Per meter
Scores comparable across meters	✅	❌
Sees a meter faulty for its whole history	✅	❌
Training rows available	~14,130	~450 per meter

The second row is decisive and not hypothetical: a per-meter model learns that meter's own history as normal, the exact blind spot Model 1's T4 hit on ZN.16-PB, whose fault covers 95.37% of readings across 18 of 19 months. A per-meter Model 3 would rank ZN.16-PB's faulted days unremarkable for eighteen months.

Decision 6 — meter identity is not a feature

An Isolation Forest scores a point by how few splits isolate it. A meter id — one-hot or ordinal — is isolable in one split, so the score drifts toward "which meter is this" and rare meters score anomalous for being rare.

Per-meter context enters as **behaviour**: `<rate>_dev28` = the rate minus that meter's trailing 28-day median, 0 with `dev28_available = False` when fewer than 14 prior days exist. Tier 2 only (Decision 12).

Decision 7 — rolling window: 180 days, refit daily, minimum 60 distinct prior dates [D]

Parameter	Value	Definition
`train_window_days`	180	training rows are all meter-days with `D − 180 ≤ day < D`
`min_train_dates`	60	**60 distinct prior Saudi calendar dates carrying observed training data** — *not* 60 meter-day rows. A single date contributing 31 meter-days counts as **one**.
Refit cadence	daily	the scored day is always out of sample

The distinction matters: 60 meter-day rows could be two calendar dates, which would calibrate a forest against essentially one moment. Sixty distinct dates guarantee the reference spans two months of facility behaviour.

Measured consequence: the first scorable Saudi date is **2025-02-07**; 60 observed dates (2024-12-09 → 2025-02-06) fall in warm-up; **506 of 566 observed dates are scorable**; and the minimum `distinct_prior_dates` among scorable days is exactly 60.

> **180 days is a pre-specified, unvalidated v1 parameter** (resolved §11.1). It will not be chosen from alternatives by maximising Model 1 overlap — that would make the cross-check circular and turn a concordance measure into a fitting objective.

Decision 8 — `contamination='auto'`, and it is not a prevalence estimate

`contamination` sets the **decision threshold** that converts scores into a binary `predict()` output. It does not change the fitted forest and does not estimate how many faults exist.

Model 3 **never calls** `predict()`. It takes `score_samples()`, ranks, and reviews the top N, so the threshold is irrelevant to every output. `contamination='auto'` is chosen so no number in the configuration can be mistaken for a prevalence claim.

This has already been misread once: the earlier prototype used `contamination=2%` and flagged 3.9% of days. Neither is an estimate of how many faulty days exist — **both are properties of a threshold applied to a ranking**, on data with no labels.

Decision 9 — score normalisation and ranking [A]

The correction. `daily_rank_pct` is a percentile *within* one Saudi day. Exactly one meter per day attains 1.0, so a global historical Top-50 taken on it would be ~506 rows all tied at 1.0 and resolved by nothing but tie-break order. It cannot rank across days and must never be used to.

Two scores are produced, with separate jobs:

Score	Comparable across	Used for
`daily_rank_pct`	within one Saudi day	within-day facility ranking, "who was worst today"
`window_anomaly_pct`	**across refits**	**global historical Top-25/50/100 and per-meter historical ranking**
`window_anomaly_z`	across refits	**deterministic secondary tie-break only**

`window_anomaly_pct` — for each daily forest `F_D`:

1. `train_scores = -F_D.score_samples(W_D)` — score the forest's own 180-day training rows, same sign convention as everything else.
2. For each meter *m* present on *D*: `raw_score = -F_D.score_samples(x_{m,D})`.
3. `window_anomaly_pct = |{t ∈ train_scores : t ≤ raw_score}| / |train_scores|`.

The result is an empirical percentile against **that forest's own reference distribution**, so a 0.98 on one date means the same thing as a 0.98 on another even though the two forests differ. Convention fixed explicitly: `≤`, so a score above every training row gives exactly 1.0.

The training scores are **in-sample**. With `max_samples=256` most training rows are out of most trees' subsamples, so the bias is small, but this is a **calibration reference, not a performance estimate**, and is recorded as such.

`window_anomaly_z` **is a tie-break, and nothing more.** `window_anomaly_pct` has a ceiling: a genuinely extreme day scores above every training row and saturates at 1.0, and several meters can saturate on the same day, reintroducing the tie problem one level up.

```
window_anomaly_z = (raw_score − mean(train_scores)) / std(train_scores)
```

> **What it is not.** It is **not** a calibrated probability, **not** a confidence, and **not** an independent anomaly measure. It is a deterministic ordering device for rows whose `window_anomaly_pct` is equal — above all at the 1.0 saturation point. Isolation Forest scores are not normally distributed, so a z-value here carries no distributional meaning and must never be read as "how many sigmas anomalous". `window_anomaly_pct` remains the primary cross-refit historical ranking score; nothing is ranked, selected or reported on `window_anomaly_z` alone.

Ranking order is `(window_anomaly_pct desc, window_anomaly_z desc, day_saudi, meter_label)` — fully deterministic, with the two label fields guaranteeing byte-identical order between runs.

Retained for audit: `raw_score`, `train_score_mean`, `train_score_std`, `n_train_rows`, `n_train_dates`, `window_start`, `window_end`, `n_meters_scored_that_day`.

Decision 10 — cross-check against Model 1, on the scored universe [B][D]

Model 1 outputs never touch the fit. Stronger than "excluded by allow-list": the fitting and scoring stage does not open a Model 1 artifact at all (§5). The join happens afterwards, into a separate cross-check artifact. Circularity is prevented by process, not by discipline.

Base rates come from the scored universe, not all meter-days. Warm-up removes 1,629 meter-days, so the preliminary denominators were wrong:

Model 1 family	base rate, all 15,759	base rate, scored 14,130	Δ
T1 polarity	19.66%	**19.61%**	−0.05
T2 current/power	8.03%	**7.83%**	−0.20
T3 power factor	48.81%	**48.80%**	−0.01
T5 register	3.84%	**3.35%**	−0.48
any family	59.67%	**58.97%**	−0.70
Critical or Problematic	50.52%	**49.58%**	−0.93

The shifts are small but they are the correct denominators, and T5 moves most because many stuck-register days sit in the commissioning-era warm-up. **These values are recomputed by the run from its own scored population and are never hard-coded**, so any future change to warm-up or scope re-derives them automatically.

The metric is **lift over base rate**, because the "any family" base rate of 58.97% means a random top-50 would "agree with Model 1" about 59% of the time:

```
overlap_f@N = share of the top-N meter-days carrying >= 1 Model 1 event of family f
lift_f@N    = overlap_f@N / base_rate_f        # 1.0 is chance
```

with a permutation null: 10,000 random N-draws **from the scored universe**, reporting the 95th percentile.

Three standing rules: agreement is **concordance, not precision** — neither model has labels; **disagreement is an output, not a failure**; and **no Model 3 parameter is ever selected by maximising lift** (§11.1, §11.3).

Decision 11 — feature ranges: only genuine rates are bounded [C]

The v1 draft asserted every feature lies in [0, 1]. That is wrong, and measurably so.

Feature group	Range	Measured
The ten **rate** features	**[0, 1]**	numerator ⊆ denominator by construction
`coverage_ratio`	**[0, ∞)**	min 0.0035, **max 2.0000, exceeds 1 on 3,173 of 15,759 meter-days (20.13%)**
`mean_load_pct_cb`	**[0, ∞)**	min 0.00, max 41.89, p99 25.90 — never exceeds 100 in this history, but **may in principle and must not be clipped**
`*_dev28`	**[−1, +1]** for rate deviations	measured −1.0000 to +0.9563, negative on 4.99% of meter-days

`coverage_ratio > 1` on a fifth of meter-days is not an anomaly — the denominator is the facility *median*, so roughly half the meters sit at or above it by definition. A [0, 1] assertion would have failed on 20% of the table for an entirely mundane reason.

The only universal constraint on the fitted matrix is that every value is finite. No feature is clipped, winsorised or rescaled: an Isolation Forest is scale-invariant, and clipping `mean_load_pct_cb` at 100 would silently erase precisely the overload condition most worth surfacing.

Decision 12 — Tier 1 is the canonical v1 fitted set (resolved §11.3)

Tier 1 alone is fitted for v1. Tier 1 + Tier 2 may be run as a **sensitivity variant**, with the ranking change reported. **No winner is chosen on Model 1 overlap.**

---

2. Exact live columns and joins

```
FROM gold.energy_raw            e
JOIN gold.meter                 m USING (energy_meter_key)
LEFT JOIN gold.zone             z USING (zone_key)
LEFT JOIN silver.dim_meter_cb_rating r USING (energy_meter_key)
GROUP BY m.meter_label, (e.reading_timestamp_utc + interval '3 hours')::date
```

Saudi-local date derived from UTC in the query. Saudi time is UTC+3 with no DST, so the Saudi day closes at `D 21:00:00 UTC`. **No Task 0A or 0B artifact is read.**

Column	Used for
`e.reading_timestamp_utc`	Saudi-date grouping, `readings_reported`
`e.active_power_total_kw`	`negative_total_power_rate`
`e.active_power_phase_1_kw` … `_3_kw`	phase reversal rate, PF load gate, current/power rates
`e.current_phase_1_a` … `_3_a`	load gate, current/power rates, `mean_load_pct_cb`
`e.power_factor_phase_1` … `_3`	gated PF-Critical rates
`e.status_amp1` … `_amp3`	`major_issue_rate`, `major_problem_rate`
`e.status_pf1` … `_pf3`	ungated PF-Critical rate (diagnostic only)
`e.status_v1` … `_v3`, `status_vn1` … `_vn3`	`voltage_not_acceptable_rate`
`e.status_hz`	`frequency_not_acceptable_rate`
`m.meter_label`	grouping key and output label (`gold.energy_raw` has none)
`z.zone_name`	reporting only, never a feature
`r.cb_max_rating_a`	the 5% load gate and `mean_load_pct_cb`

Not used: `active_energy_cumulative_kwh` and anything derived from it, and the eight columns verified empty over the full history.

Verified: `status_amp*` **and** `status_pf*` **are NULL on zero readings**; `active_power_total_kw` is present on all 4,535,918; every meter has a `cb_max_rating_a`. No denominator needs a NULL guard.

---

3. Why rates rather than counts — the measured case

Period	Median readings/day
2024-12 → 2026-03	286–287
from **2026-04-11**	309–315
2026-07	**354**

A **23.8% shift**, plus per-day variation from 1 reading to 354. A count feature inherits all of it: the same condition yields a 24% larger count after 2026-04-11 purely because the meter reports more often, and the forest would learn the regime change as an anomaly.

Honestly stated: within the *current* data a raw count correlates only **+0.021** with cadence, because cadence is nearly constant on most days. The case for rates rests on the **temporal regime change** and on **partial days**, not on a large cross-sectional correlation — and being structural, it holds regardless.

---

4. Exact feature definitions

Let `R` = `readings_reported`, and for phase *k*:

```
gated_k = count of readings where current_phase_k_a >= 0.05 * cb_max_rating_a
                                 AND active_power_phase_k_kw > 0
```

The 5% gate is the production classifier's own `third = margin / 3 = 0.05 * CB`. No new constant is introduced.

Tier 1 — canonical v1 fitted set

#	Feature	Numerator	Denominator	Bounded [0,1]?
1	`coverage_ratio`	`R`	facility median `R` that date	**no**
2	`major_issue_rate`	Σ_k `status_amp_k = 'Major Issue'`	`3R`	yes
3	`major_problem_rate`	Σ_k `status_amp_k = 'Major Problem'`	`3R`	yes
4–6	`pf_critical_rate_gated_k`	phase *k* gated **and** `0 <= pf_k < 0.6`	`gated_k`	yes
7–9	`pf_gate_open_rate_k`	`gated_k`	`R`	yes
10	`negative_total_power_rate`	`active_power_total_kw < 0`	`R`	yes
11	`negative_phase_power_rate`	Σ_k `active_power_phase_k_kw < 0`	`3R`	yes
12	`current_without_power_rate`	Σ_k `current_k > 0 AND kw_k = 0`	`3R`	yes

#	Feature	Numerator	Denominator	Bounded [0,1]?
13	`power_without_current_rate`	Σ_k `current_k = 0 AND kw_k <> 0`	`3R`	yes
14	`voltage_not_acceptable_rate`	Σ over 6 voltage statuses `<> 'Acceptable'`	`6R`	yes
15	`frequency_not_acceptable_rate`	`status_hz <> 'Acceptable'`	`R`	yes
16	`mean_load_pct_cb`	mean over readings of `(Σ_k current_k)/3`	`cb_max_rating_a`, ×100	**no**

Features 7–9 exist for a specific reason. When `gated_k = 0` the PF rate is **undefined**, and an Isolation Forest cannot take NaN. Imputing 0 would silently assert "power factor was fine" when the truth is "we never had enough load to judge". Carrying `pf_gate_open_rate_k` alongside distinguishes *rate 0 because PF was good* from *rate 0 because the gate never opened*. This is not a corner case: **4,262 of 15,759 meter-days (27.04%) never open the phase-1 gate.** Convention: `gated_k = 0` ⇒ both are `0.0`, and the pair is unambiguous.

Features 14–15 are retained for v1 (resolved §11.5) despite being all but constant — measured, **12 non-acceptable of 27,215,508 voltage classifications and 1 of 4,535,918 frequency classifications**, 13 in 31,751,426. They cost two columns and would be the only signal if a voltage problem ever appeared.

Tier 2 — per-meter deviation, sensitivity variant only

`*_dev28` for features 2, 3, 4–6, 10, 11, plus `dev28_available`. Nine columns. Not in the canonical v1 fit.

Diagnostic columns, never fitted

`pf_critical_rate_ungated_k` (three) so the gated-vs-ungated comparison in §1 is reproducible on any future run and a reviewer can see why a day was *not* flagged.

---

5. Artifacts and schema [B]

Model 1 is kept out of the Model 3 pipeline by **separating the artifacts**, not merely by column selection. The fit-and-score stage never opens a Model 1 file.

#	Artifact	Contents	Model 1?
1	`model3_feature_table.parquet`	raw Gold-derived features only	**none**
2	`model3_scores.parquet`	`daily_rank_pct`, `window_anomaly_pct`, `window_anomaly_z`, `raw_score`, window provenance	**none**
3	`model3_model1_crosscheck.csv`	artifact 2 joined to the frozen Model 1 events, base rates, lift, permutation null	produced **after** scoring
4	`model3_review_lists.csv`	lists A, B and C (§8)	C needs artifact 3
5	`model3_coverage_report.csv`	coverage and scored-universe distributions	none

`model3_feature_table.parquet`

Column	Type	Role
`energy_meter_key`	str (uuid)	key
`meter_label`, `short_label`	str	key / label
`zone_name`, `is_zone13`	str / bool	reporting
`cb_max_rating_a`	float	reporting; builds the gate
`day_saudi`	date	key
`readings_reported`	int	provenance
`gated_readings_phase_1..3`	int	provenance, the PF denominators
Tier 1 features 1–16	float	**fitted**
Tier 2, 9 columns	float / bool	fitted in the sensitivity variant only
`pf_critical_rate_ungated_1..3`	float	diagnostic, never fitted
`low_coverage_day`	bool	reporting, never a filter
`scorable`	bool	passes the warm-up rule (Decision 7)

Row key `(energy_meter_key, day_saudi)`, unique. **15,759 rows** at current history, of which **14,130 are** `scorable`. No `m1_*` column exists in this file — asserted by test.

---

6. Row counts, coverage and the scored universe

Observed meter-days	**15,759**
Meters	**31** (2 in Zone 13)
Calendar span	2024-12-09 → 2026-07-11 (580 days)
Distinct Saudi dates with any data	**566** (14 dates have none)
Complete grid would be	17,980
Absent meter-days	**2,221**
Warm-up dates excluded	**60** (2024-12-09 → 2025-02-06)
First scorable Saudi date	**2025-02-07**
Scorable dates	**506 of 566**
Model 3 scored universe	**14,130 meter-days**

Readings per meter-day

percentile	1%	5%	25%	50%	75%	95%	99%	max
all meters	157	269	286	287	287	316	354	354
Zone 13 only	204	268	269	269	270	299	354	354

Low-coverage days: < 10 readings on 35 meter-days (0.22%), < 50 on 74 (0.47%), < 100 on 114 (0.72%), < 200 on 182 (1.15%), < 250 on 292 (1.85%).

Candidate rate distributions

feature	mean	median	p90	non-zero days
`major_issue_rate`	0.0027	0.0000	0.0000	1.57%
`major_problem_rate`	0.0034	0.0000	0.0000	6.66%
`pf_critical_rate_1` **ungated**	0.3328	0.0185	1.0000	**54.16%**
`pf_critical_rate_1` **gated**	0.0402	0.0000	0.0392	**12.84%**
`negative_total_power_rate`	0.0621	0.0000	0.0209	11.49%

The ungated PF rate correlates **−0.648** with `pf_gate_open_rate_1`. A feature that moves inversely with load is measuring load, not power quality. Gating cuts non-zero days from 54.16% to 12.84% and is the single most consequential choice in this design.

---

7. Isolation Forest fitting strategy

Setting	Proposed	Note
Scope	one **global** forest per scoring day	Decision 5
Training rows	preceding 180 days, all meters	strictly before the scored day
Warm-up rule	≥ 60 **distinct prior Saudi dates**	Decision 7
`n_estimators`	200	fixed in advance, not tuned
`max_samples`	256	library default and the original paper's recommendation
`contamination`	`'auto'`	never read as prevalence; `predict()` never called
`random_state`	fixed	required for stable ranking
Features	Tier 1 by allow-list	Tier 2 as a separate sensitivity variant
Scaling	none	scale-invariant; clipping would erase real overload

Procedure for Saudi day *D*:

1. Assemble training rows for `[D − 180, D − 1]`; require ≥ 60 distinct dates or skip *D*.
2. Fit one forest `F_D`.
3. `train_scores = −F_D.score_samples(training rows)` — the calibration reference.
4. Score the meters present on *D* — out of sample by construction.
5. Emit `daily_rank_pct`, `window_anomaly_pct`, `window_anomaly_z`, `raw_score` and the window provenance.

Nothing is fitted yet. Parameters are a proposal for review.

---

8. Review lists and the Model 1 overlap metric

List	Definition	Ranked on	Size
A — facility	most unusual meter-days over the whole history	`window_anomaly_pct` **[A]**	N ∈ {25, 50, 100}, default 50

List	Definition	Ranked on	Size
B — per meter	each meter's own most unusual days	`window_anomaly_pct`	2 per meter (62)
C — unexplained	list A entries where **the same** `(meter_label, day_saudi)` carries no relevant Model 1 electrical event	—	variable, reported separately **(§11.4)**

N = 50 is a **historical reporting default only**; operational review capacity is TBD (§11.2). All three lists carry the standing caveat that these are review candidates.

List C is produced as its own artifact and is never folded into A. It is the only output that can surface something no deterministic rule covers, and its review quota is TBD.

What "unexplained" means, exactly

"Explained" is judged **per meter-day, never facility-wide**: a Model 3 item is explained only if *that same meter* carried a relevant Model 1 event on *that same Saudi day*. Another meter having an event that day explains nothing about this one.

A Model 1 event counts as explaining an electrical anomaly only when **all** hold:

Condition	Reason
Same `(meter_label, day_saudi)`	a facility-wide reading would mark almost everything explained
`event_class = 'ELECTRICAL_FAULT'`	the event asserts an electrical condition
`commissioning = False`	a commissioning artifact is not an explanation

`COMMISSIONING` and `DATA_AVAILABILITY` events are **still joined and still shown** as context columns on every cross-check row, because a reviewer needs to know a day was an outage or a commissioning window. They simply never satisfy "Model 1 explains this". A day whose only Model 1 event is a telemetry outage is therefore **unexplained**, which is the correct reading: the outage says why data is thin, not why the electrical behaviour was unusual.

Overlap metric

Per family and for Critical/Problematic only, at N ∈ {25, 50, 100}:

Quantity	Definition
`base_rate_f`	share of the **scored universe** with ≥ 1 Model 1 event of family *f*
`overlap_f@N`	share of list A with ≥ 1 such event
`lift_f@N`	`overlap_f@N / base_rate_f` — **1.0 is chance**
`null_p95_f@N`	95th percentile of `overlap_f@N` over 10,000 random N-draws from the scored universe
`unexplained@N`	size of list C

The headline is **lift**, never raw overlap.

---

9. Tests

Cadence normalisation

Test	Criterion
A meter-day duplicated to twice the reading count, same conditions, gives **identical** rates	every Tier 1 rate equal to within 1e-12
Doubling reading count changes only `coverage_ratio` and `readings_reported`	all other features unchanged
A raw-count version of the same feature **does** change	proves the test has teeth
The 2026-04-11 cadence regime change produces no step in any rate's facility median	median shift < 1 standard deviation

Feature ranges [C]

Test	Criterion
The ten **rate** features lie in [0, 1]	enumerated by name, not by "all features"
`coverage_ratio` is **allowed** to exceed 1	a fixture above 1 passes validation; measured on 20.13% of real rows
`mean_load_pct_cb` is **allowed** to exceed 100 and is **not clipped**	a 150% fixture survives unmodified into the fitted matrix
`*_dev28` is **allowed** to be negative	a −0.4 fixture passes
Every fitted value is **finite**	no NaN, no ±inf — the one universal constraint, asserted before fitting
No feature is winsorised or rescaled	fitted matrix equals the table's own values

Missing-day behaviour

Test	Criterion
A meter-day with no readings produces **no row**	absent from the table
Days before a meter's first reading and after its last are excluded	row count equals observed meter-days
Interior gap days are counted in metadata	`missing_interior_days` non-zero
A 4-reading day is **retained**	present, not filtered
`gated_k = 0` ⇒ `pf_critical_rate_gated_k = 0` **and** `pf_gate_open_rate_k = 0`	unambiguous pair, no NaN reaches the fit

Saudi-date grouping

Test	Criterion
`2026-03-01 20:59:59 UTC` → Saudi day 2026-03-01; `21:00:00 UTC` → 2026-03-02	exact boundary
Every Saudi day spans 86,400 seconds	no DST in the history
SQL grouping and pandas grouping agree on every meter-day	identical for all 15,759 rows

Warm-up and the scored universe [D]

Test	Criterion
`min_train_dates` counts **distinct Saudi dates**, not meter-day rows	a window of 2 dates × 31 meters = 62 rows is **not** scorable
First scorable date is 2025-02-07 with the configured window	exact
Scored universe is 14,130 meter-days	exact, asserted
Base rates are computed from the scored universe, never from all 15,759	recomputed in-run, not hard-coded
The permutation null draws from the scored universe only	sampled population equals the scored population

Ranking [A]

Test	Criterion
`daily_rank_pct` is a permutation within each day, in [0, 1]	no gaps or duplicates per day
Exactly one meter per day attains `daily_rank_pct = 1.0`	demonstrates it cannot rank globally
A global Top-N by `window_anomaly_pct` **is NOT the set of daily winners**	the two orderings differ on a fixture where every day has a rank-1.0 meter
Global Top-N is not determined by daily-rank ties	on a fixture where 50 days each have a 1.0 daily rank, the Top-50 by `window_anomaly_pct` selects a different set, ordered by score not by date
`window_anomaly_pct` is comparable across refits	two forests with different windows give the same percentile for the same relative position in their own training distributions
`window_anomaly_pct` uses `≤`	a score equal to the maximum training score gives exactly 1.0
`window_anomaly_z` breaks ceiling ties	two meters at `window_anomaly_pct = 1.0` order by z, deterministically
`raw_score` orientation is correct — a synthetic extreme outlier ranks **first**	catches the sklearn sign convention
Two runs on identical input give byte-identical ranks	full-table equality
Ties resolve by `(day_saudi, meter_label)`	synthetic ties resolve identically across runs

Scope and independence [B]

Test	Criterion
`model3_feature_table.parquet` contains **no** `m1_*` column	column-name assertion
`model3_scores.parquet` contains **no** Model 1 column	column-name assertion
The fit-and-score stage opens **no** Model 1 artifact	no read of any `model1_*` path during fit/score
The cross-check artifact is produced only **after** scoring	ordering enforced by the code path
No Task 0A/0B artifact is read	nothing outside `gold.*` and `silver.dim_meter_cb_rating`

Test	Criterion
No register-derived column is used	`active_energy_cumulative_kwh` absent from the feature list
Read-only, REPEATABLE READ verified before any query	connection facts recorded
Row count equals 15,759, scored 14,130, both asserted and failing loudly	scope assertion

10. Definition of Done

- [ ] `src/model3_triage/features.py` — daily feature table, rates only, from `gold.energy_raw` + `gold.meter`, no Task 0A/0B dependency, **no Model 1 read**
- [ ] `config/model3_config.yaml` — every threshold, window and parameter; none in code
- [ ] `outputs/<ts>/model3_feature_table.parquet` — 15,759 rows, 14,130 `scorable`, §5 schema
- [ ] `outputs/<ts>/model3_coverage_report.csv` — §6 distributions and the scored universe
- [ ] `tests/test_model3.py` — every test in §9 passing
- [ ] `run_metadata.json` — connection facts, scope assertion, config fingerprint, the explicit statement that `contamination` is not a prevalence estimate, that `window_anomaly_pct` is calibrated in-sample, and that no precision or recall is claimed
- [ ] No Isolation Forest fitted, no score produced, no review list emitted

Fitting, scoring, the review lists and the overlap metric are the **next** stage.

11. Resolved decisions, 2026-09-20

#	Question	Resolution
1	180-day window	**Pre-specified, unvalidated v1 parameter.** Not chosen among alternatives by maximising Model 1 overlap.
2	N = 50	**Historical reporting default only.** N = 25/50/100 all reported; operational capacity TBD.
3	Tier 2 deviation features	**Tier 1 is canonical v1.** Tier 1+2 may run as a sensitivity variant; no winner chosen on overlap.
4	List C review budget	**Produced separately and kept visible.** It holds candidate findings Model 1 does not explain; quota TBD.
5	Voltage and frequency features	**Kept in v1.**

Corrections applied in v2

	Correction	Where
A	`daily_rank_pct` cannot rank across days; added `window_anomaly_pct` (+ `window_anomaly_z` for the ceiling)	Decision 9, §7, §8, §9
B	Model 1 excluded from the feature and score artifacts entirely; joined only afterwards	Decision 10, §5, §9

	Correction	Where
C	Only genuine rates are bounded in [0, 1]; `coverage_ratio`, `mean_load_pct_cb` and `*_dev28` are not, and nothing is clipped	Decision 11, §4, §9
D	Base rates recomputed on the 14,130-meter-day scored universe; `min_train_dates` defined as distinct Saudi dates	Decision 7, Decision 10, §6, §9

---

12. Implementation record, 2026-09-20

Implemented in the approved sequence. Model 3 is **v1 complete**.

Stage	Artifact	Model 1 opened?
1 — features	`model3_features_20260920T083853Z/`	**no**
2 — fit and score	`model3_20260920T085615Z/model3_scores.parquet`	**no**
3 — cross-check	`model3_20260920T085615Z/model3_model1_crosscheck.parquet`	yes, after all scoring

Reconciliation, all 8 exact: 4,535,918 readings (total and re-summed from the daily table), 31 meters, 15,759 meter-days, 566 distinct Saudi dates, 14,130 scored, first scorable date 2025-02-07, row key unique. Warm-up removed 60 dates / 1,629 meter-days.

Correction A is confirmed empirically. `daily_rank_pct == 1.0` on exactly **506** rows — one per scored date — so a historical Top-50 taken on it would have drawn from 506 tied values. `window_anomaly_pct` has **13,934 distinct values** across 14,130 rows and **8 saturate at 1.0**; `window_anomaly_z` orders those eight (4.25 → 3.63). The ceiling was real and small, and the tie-break handles it.

Determinism: a full refit from scratch produced byte-identical frames, `max |raw_score delta| = 0.000e+00`.

Three implementation bugs, each found by a test or a run

1. **Ambiguous join.** `zone_key` exists on both `gold.energy_raw` and `gold.meter`, so `USING (zone_key)` raised `AmbiguousColumn`. Joined explicitly through `gold.meter`.
2. **Label/position index mismatch.** `groupby(...).index` returns pandas *labels* while the feature matrix is indexed *positionally*. These coincide for a clean `RangeIndex` from parquet, so the full run worked by accident; any filtered input raised `IndexError` or — worse — would have silently selected the wrong training rows. `score_all_days` now resets the index and uses positions throughout.
3. **Object-dtype boolean.** `merge(...).fillna(False)` leaves an *object* column, and `~` on an object Series of Python bools performs integer bitwise negation (`True → −2`) rather than logical NOT, which broke List C with a `KeyError`. All join flags are now cast with `.astype(bool)` and a regression test asserts `~flag` is logical.

Bug 2 is the one worth remembering: it produced correct output on the real data and would only have surfaced later, on a subset or a re-indexed frame.

What the results say

Model 3's top of list is concentrated on the **intermittent** polarity meters — ZN.11-PB.2, ZN.10-PB.2, ZN.10-PB.1 and ZN.12-PB take 84 of the Top-100 — and largely not on the chronic ones (ZN.16-PB 5, ZN.8-PB.2 2, ZN.5-PB 1). That is the expected consequence of Decision 5: a constant fault sits in the training window every day and is

therefore *normal* to the forest, while an intermittent fault produces genuinely unusual days.

Model 1 explains **99 of the Top-100**. As a triage layer Model 3 is therefore adding ordering within days Model 1 already flags, rather than surfacing much that Model 1 misses. T5 lift is **1.19 with an empirical p-value of 0.43–0.57** — indistinguishable from chance, so Model 3 does not preferentially surface register-integrity days.

No precision, recall, accuracy or confirmed-fault detection is claimed anywhere.